MEER FOUNDATION | EDUSUTO — CASE STUDY

Case Study 4 of 6 · Module 3 & 5

Community Biochar Carbon Project — South Asian Smallholders

Aggregating Hundreds of Smallholder Farmers for Scalable Agricultural CDR

QUICK FACTS

Sector: Agricultural Carbon Dioxide Removal (CDR)

Location: Multiple districts, India and Bangladesh (composite case)

Technology: TLUD kilns and cone kilns; VM0042 (Verra); Gold Standard soil carbon

1. Background: The Smallholder Carbon Opportunity

India's 150 million farm households represent one of the largest untapped carbon sequestration opportunities in the world. Average smallholder farm size in India is 1.08 hectares — too small for individual carbon project development, but collectively enormous. The challenge has always been aggregation: how to group hundreds or thousands of tiny farms into a single, verifiable carbon project that meets international standards.

This composite case study draws on multiple real smallholder biochar projects across South Asia — including projects registered with Verra (VM0042) and Gold Standard in India and Bangladesh — to illustrate the operational model, financial structure, and key challenges of this project type.

Project type	Smallholder biochar production and soil application (VM0042)
Scale	500 farmers across 3 districts; total area ~600 hectares
Implementing organisation	Farmer Producer Organisation (FPO) with NGO technical partner
Feedstock	Rice straw (primary), wheat straw, cotton stalks (seasonal)
Technology	TLUD kilns (household scale, 200L); cone kilns (community scale)
Annual biochar production	~1,050 tonnes/year (Year 3 steady state)
Annual CO■e sequestered	~2,600–2,800 tCO■e/year (after buffer deduction)
Carbon standard	Verra VCS + VM0042 + CCBS (Gold Level)
Credit price	$28–35/tCO■e (voluntary market, CCBS premium)
Farmer revenue	■1,200–1,800 per farmer per year from carbon credits
Project term	10 years (crediting period)

2. Project Design: The Aggregation Model

The defining challenge of a smallholder carbon project is aggregation. Individual farmers cannot afford project development costs ($50,000–150,000 for registration alone). The solution is to develop a grouped project under which multiple farmers participate under a single Project Design Document (PDD).

Legal Structure:

- The FPO (Farmer Producer Organisation) acts as the legal project proponent, holding the registry account and signing the offtake agreement with buyers. Individual farmers sign a Farmer Participation Agreement with the FPO, committing to the required practices for the project term.
- The NGO technical partner provides training, monitoring, data management, and technical oversight. It receives a service fee from carbon revenues.
- A Development Finance Institution (DFI) or impact investor provides upfront project development capital (bridge financing), which is recovered from the first two to three years of credit sales.

Benefit Sharing Framework:

Revenue Recipient	Share	Purpose
Farmers (direct)	60%	In-kind kiln subsidy (first year) + annual cash payment
Community fund	5%	Village-level infrastructure: water, solar lights
FPO operations	15%	FPO staff, field officers, coordination
NGO technical partner	10%	Training, DMRV, data management
Project development cost recovery	10%	Repayment of upfront capital over 3–4 years

3. Operations: From Crop Residue to Carbon Credit

Phase	Activity	Who Does It	Key Output
Enrollment	Baseline SOC sampling, farmer consent, GPS mapping	Field officers + lab	PDD baseline data; enrolled farmer records
Training	Kiln operation, feedstock preparation, safety, record-keeping	NGO trainer	Certified farmer operators
Production (Year-round)	Biochar production from crop residue; GPS-tagged batch records	Farmer	Monthly batch logs; biochar inventory
Soil application	Charged biochar applied to fields at recommended rate	Farmer + field officer	Application records; GPS photos

Phase	Activity	Who Does It	Key Output
Monitoring (Annual)	SOC sampling; activity data audit; satellite cross-check	Field officers + lab	Annual monitoring report
Verification (Bi-annual)	Third-party audit of data; site visits; sample re-testing	Accredited VVB	Verification report
Issuance	Credits issued to FPO registry account	Verra	Carbon credits (VCUs)
Sale	Credits sold to buyer per offtake agreement	FPO + broker	Cash to FPO account
Distribution	Revenue distributed per benefit sharing framework	FPO	Farmer payments + community fund

4. DMRV: Digital Tools at Work

- **Mobile data collection:** Field officers use KoboToolbox on smartphones to capture batch data (feedstock weight, biochar weight, kiln temperature, GPS location) at each production event. Data syncs to a cloud database nightly.
- **Satellite monitoring:** Sentinel-2 satellite imagery (10m resolution, freely available) is used to cross-check field boundaries, monitor crop growth as a proxy for practice adoption, and flag anomalies requiring ground-truthing.
- **IoT temperature logging:** Low-cost data loggers (approx. ■800 each) are attached to kilns to automatically record peak pyrolysis temperature at each production event — confirming that temperatures sufficient for Class B biochar (H:Corg $\leq$ 0.6) are achieved.
- **Blockchain registry:** Each batch of biochar is assigned a unique digital ID linked to its production data. This creates a traceable chain from crop residue input to soil application to carbon credit, visible to buyers and verifiers.

5. Financial Analysis: Does It Work?

Metric	Value (Year 3 Steady State)
Annual gross credits	2,730 tCO■e
After 15% buffer deduction	2,321 tCO■e
Credit price	$30/tCO■e (CCBS premium)
Gross carbon revenue	~■58 lakh (~USD 69,600)
Less platform fees (8%)	~■4.6 lakh
Net distributable revenue	~■53.4 lakh
Farmer share (60%)	~■32 lakh = ~■6,400 per farmer

Metric	Value (Year 3 Steady State)
Project operating cost	~■21 lakh/year
Net project margin	~■0.4 lakh/year (Year 3; improving with scale)
10-year farmer income (NPV at 8%)	~■42,000 per farmer (cumulative)
Break-even year	Year 2 (after capital recovery)

The INR 6,400/year carbon income per farmer is modest but meaningful — representing 8–12% of average annual farm income in the target districts. Combined with yield improvement benefits (■3,000–5,000/year value from reduced fertiliser use and better soil water retention), total project benefit per farmer is approximately ■9,000–11,000/year.

6. Challenges and How They Were Addressed

Challenge	What Happened	Solution
Farmer dropout (Year 1–2)	15% of enrolled farmers stopped producing biochar due to time constraints	Simplified production protocol; group production sessions; peer champion model
Feedstock seasonal gaps	No rice straw available June–September	Alternative feedstocks trained: cotton stalk, maize cob, wood prunings
Lab sample transport	Samples degrading before reaching lab 400 km away	Partnered with state agricultural university laboratory 60 km away
Buyer price fluctuation	Spot market prices fell 18% in Year 2	Secured 5-year floor price offtake agreement with corporate buyer before Year 3
Fraudulent batch records	2 farmers found to have inflated batch weights	Random audit protocol (20% of batches physically cross-checked); farmer re-training

7. Lessons for Indian Carbon Project Developers

- **FPO partnership is essential:** Projects run through existing FPOs have 60–70% lower farmer acquisition costs and better retention rates than standalone NGO-run projects.
- **Floor price offtake is non-negotiable:** Carbon revenue volatility is the single biggest risk to smallholder biochar projects. Securing a floor price before launch protects farmers and enables financial planning.
- **DMRV reduces but does not eliminate field visits:** Digital tools reduce monitoring cost by 40–60%, but physical verification visits remain essential for buyer confidence and audit compliance.

- **Scale for viability:** Projects below 300 farmers struggle to cover development costs. The economic sweet spot is 500–2,000 farmers at steady state.
